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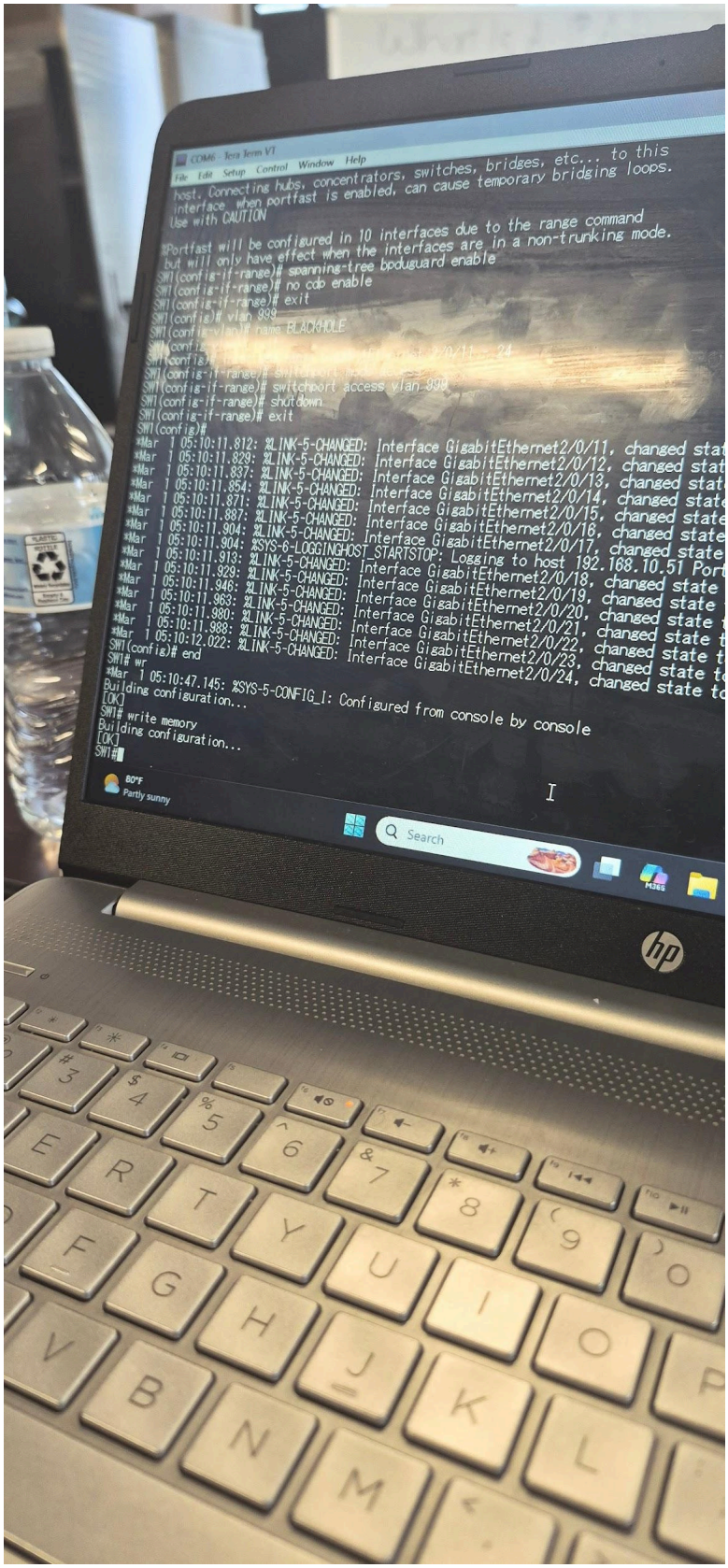
Setup Lan Paragraphs

First to setup the Lan using the switch with Teraterm, in (**step 1)** we have to plug in the light blue ethernet cable into the switch and usb into the laptop. It has a power switch as well and then we launch Teraterm. (**Step 2)** Wiping the old data and settings is key since there has been a lot of work done on it before. (**Step 3)** An IP address is then made for the switch.

So departments in a business don't have access to each other's data, (**Step 4)** we will create different VLAN numbers for each department e.g Marketing, Sales, Administration etc. (**Step 5)** Instead of FastEthernet ports, we applied restrictions on our GigabitEthernet ports. (**Step 6)** After this, we issue a warning that an unauthorized person trying to access the switch will be prosecuted. (**Step 7)** For managing the switch remotely, we'll create hostnames, local user accounts and password. If someone wants to access the switch without authorization, they won't be able to see what they type in the terminal. (**Step 8)** We'll create 5 virtual interfaces (vlan 10, vlan 20, vlan 30 etc).

(**Step 9)** We then made sure the Sales department VLan can't talk to the Executives VLAN. This is done by denying their access with the Sales Vlan number being 30. (**Step 10)** Configuring DHCPs is next and with that comes creating the DHCP addresses, dns server. (**Step 11)** After this, we assign physical ports for each their department with their respective VLAN. A major thing when configuring the DHCP server is to match their VLAN with their department pool. If not, you'll run into issues when looking for their IP address in step 11. You can verify this in the command prompt. (**Step 12),** we back the switch configuration to a TFTP server with the filename being a text file. Firewalls and security software could be a hindrance. (**Step 13),** ensuring best optimization for each interface range is last.

IMAGE is on Second Page


COM6 - Tera Term VT
File Edit Setup Control Window Help
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast will be configured in 10 interfaces due to the range command
but will only have effect when the interfaces are in a non-trunking mode.
SW1(config-if-range)# spanning-tree bpduguard enable
SW1(config-if-range)# no cdp enable
SW1(config-if-range)# exit
SW1(config)# vlan 999
SW1(config-vlan)# name BLACKHOLE
SW1(config-if-range)# switchport access vlan 999
SW1(config-if-range)# shutdown
SW1(config-if-range)# exit
SW1(config)#
*Mar 1 05:10:11.812: %LINK-5-CHANGED: Interface GigabitEthernet2/0/11, changed stat
*Mar 1 05:10:11.829: %LINK-5-CHANGED: Interface GigabitEthernet2/0/12, changed stat
*Mar 1 05:10:11.837: %LINK-5-CHANGED: Interface GigabitEthernet2/0/13, changed stat
*Mar 1 05:10:11.854: %LINK-5-CHANGED: Interface GigabitEthernet2/0/14, changed state
*Mar 1 05:10:11.871: %LINK-5-CHANGED: Interface GigabitEthernet2/0/15, changed state
*Mar 1 05:10:11.887: %LINK-5-CHANGED: Interface GigabitEthernet2/0/16, changed state
*Mar 1 05:10:11.904: %LINK-5-CHANGED: Interface GigabitEthernet2/0/17, changed state
*Mar 1 05:10:11.904: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.10.51 Port
*Mar 1 05:10:11.913: %LINK-5-CHANGED: Interface GigabitEthernet2/0/18, changed state
*Mar 1 05:10:11.929: %LINK-5-CHANGED: Interface GigabitEthernet2/0/19, changed state
*Mar 1 05:10:11.946: %LINK-5-CHANGED: Interface GigabitEthernet2/0/20, changed state
*Mar 1 05:10:11.963: %LINK-5-CHANGED: Interface GigabitEthernet2/0/21, changed state
*Mar 1 05:10:11.980: %LINK-5-CHANGED: Interface GigabitEthernet2/0/22, changed state
*Mar 1 05:10:11.988: %LINK-5-CHANGED: Interface GigabitEthernet2/0/23, changed state
*Mar 1 05:10:12.022: %LINK-5-CHANGED: Interface GigabitEthernet2/0/24, changed state
SW1(config)# end
SW1# wr
*Mar 1 05:10:47.145: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
SW1# write memory
Building configuration...
[OK]
SW1#
80°F
Partly sunny
Search